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OVERVIEW

The characteristics of seminal fluid and the forensic tests available to identify it

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Abstract

Semen body fluid identification remains a critical aspect of the forensic investigation of alleged sexual assaults. Over the years, as advancements have been made in the area of forensic science, several different methods for semen identification have been utilized. It is not unusual for several methods of semen identification to be incorporated into a single sexual assault investigation. Most methods of semen identification are based on either detecting enzymatic activities or are targeted at specific cells or antigens. Other methods have evolved utilizing fluorescence or protein expression. Presumptive tests are often used to screen or locate semen staining. Further confirmatory tests can be used to identify the staining as semen and finally the donor of the stain can be further identified through DNA analysis. For routine application of these methods, they need to be specific, sensitive, cost effective, easy to use, and nondestructive. Although new techniques are being researched, it appears that traditional methods of semen identification remain the most widely applied in forensic casework.

This article is categorized under:
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KEYWORDS

body fluid identification, forensic, semen, seminal fluid, sexual assault

1 | INTRODUCTION

In the forensic investigation of sexual assault allegations, it is often crucial to determine whether or not semen is present. Regardless of the other evidence that may be available, or the various scenarios that may be presented to explain them, the detection of semen provides irrefutable proof of a sexual element to an alleged assault. Although the presence of semen is not always essential to confirm the sexual element of a crime, once located it is likely to influence the progress and the outcome of an investigation.

The use of scientific screening methods to test for semen in sexual assault cases is first described in the late 19th century. Cutting-edge technology at the time allowed scientists to microscopically detect spermatozoa in suspected semen samples; however, detecting or locating the appropriate samples to perform this testing on was fraught with complications and limited its usefulness in many cases. The types of samples a forensic scientist may be presented with in a sexual assault case will likely dictate what tests are appropriate and applicable. In many cases, a medical examination of the complainant will provide a range of intimate and targeted samples for testing. There may however be circumstances of the case that require examination for the presence of dried seminal fluid staining on items such as underwear, bedding, or other garments worn by either

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complainant or accused. The physical location of the alleged assault may also be examined which can include a wide variety of surfaces that need to be screened for the presence of semen, from carpet to car seats or lawn to bare earth.

Over the years, a broad range of chemical and biochemical markers in semen have been investigated in efforts to develop methods to be able to locate and confirm the presence of semen. In 1984, Gaensslen (Gaensslen, 1984) produced a sourcebook of forensic serology which includes a detailed and comprehensive discussion of the history and development of body fluid identification with respect to semen and vaginal secretions. Beyond this period, we entered an era of rapid expansion of available forensic techniques, especially with the advent of DNA profiling. Corresponding with this expansion was a flurry of research on catalytic, enzymatic, and immunological tests for biological fluids. This research has gradually been improved and refined to produce a well-established and robust suite of techniques for the detection of semen; however, with the continual advancement of scientific knowledge and modern technology, there are new concepts and methodologies being tested to develop new approaches for age-old techniques.

2 | ANATOMY AND PHYSIOLOGY

2.1 | Anatomy of the male reproductive system

Before entering a discussion on the detection of semen in any depth, it is worth spending some time to develop an understanding of the anatomical components involved, their function and contribution to the male ejaculate.

The primary function of the male reproductive system is to produce and deliver spermatozoa, the male gametes, as a component of semen, into the female reproductive tract, to facilitate fertilization. The female reproductive system has the complementary role of producing ova, the female gamete, and delivering them, via the uterine tubes where fertilization typically occurs, to the uterus. Post fertilization, or arguably post ejaculation, the male reproductive role is completed.

The male reproductive system is comprised of the testes, the sex accessory ducts, and sex accessory glands. Each of the testes resides within the scrotum and is ovoid, approximately 4 cm long and 2.5 cm in diameter. Each teste is protected or sheathed by two layers, the *tunis vaginalis* as the outer layer and the *tunis albuginea* as an inner layer. The *tunis albuginea* extends into the testes dividing it into approximately 250–300 separate wedge-shaped compartments. Each of these compartments, also known as lobules, contains one to four tightly coiled seminiferous tubules.

The seminiferous tubules are where spermatozoa are produced. The walls of the seminiferous tubules are lined with male germ cells and sertoli cells. The male germ cells continually undergo spermatogenesis during which the diploid male germ cells each replicate (mitosis) and divide (meiosis) to produce four haploid spermatid cells. These spermatid cells then undergo spermiogenesis and are transformed into immature spermatozoa.

The time taken for the entire process of spermatogenesis can take anywhere from 64 to 120 days, depending on the method of measurement. On any single day, a male will produce in the range of 200–400 million spermatozoa, which can equate to the production of approximately 1,000 spermatozoa per minute. The seminiferous tubules within each lobule converge into a tubular network located at the rear of the testes which directs the accumulating spermatozoa into the epididymis.

Spermarche is the male equivalent of menarche and represents the pubertal changes when the testes begin to produce spermatozoa. The age at which puberty begins can be affected by genetic and environmental factors, but is generally considered to be around 10–13 years of age.

The epididymis is the first section of the sex accessory ducts through which spermatozoa will travel to the outside of the body. The epididymis is described as a comma-shaped structure at the rear of the testes, measuring approximately 3.8-cm long and containing three sections. The sections are referred to as the head (*caput)*, body (*corpus*), and tail (*cauda*). The role of the epididymis is to nurture the sperm into maturity and to concentrate and store them. The transit through the epididymis can take around 2 weeks and the now mature spermatozoa can be stored in the tail of the epididymis for about a month before they begin to be broken down and reabsorbed by the body.

The tail of the epididymis merges into the vas deferens, the second sex accessory duct. The vas deferens is approximately 45-cm long and ascends from the rear of each teste travels through the inguinal canal into the pelvic cavity where they loop over the urinary bladder. The end of the vas deferens expands slightly into a portion known as the ampulla, which can also serve as a reservoir for spermatozoa. The ampulla joins with the duct from a seminal vesicle to form an ejaculatory duct which then passes through the prostate gland to join with the urethra, the third and final sex accessory duct.

The seminal vesicles, the prostate and the bulbourethral glands are the sex accessory glands. The seminal vesicles are approximately 5-cm long and reside on either side of the bladder against the rear wall. They produce the majority of seminal plasma and are responsible for approximately 60% of the volume of semen.

The urethra extends from the bladder through the prostate and along the length of the penis to the opening in the head of the penis. The urethra serves as the route for urine and semen to exit the penis. The internal urethral sphincter is a sphincter muscle surrounding the urethra between the bladder and the prostate. This is an involuntary sphincter which remains closed when not passing urine.

The portion of the urethra that runs through the prostate is known as the prostatic urethra. The external urethral sphincter is a muscle surrounding the membranous urethra situated between the prostate and the penis within the urogenital diaphragm. This sphincter has a voluntary response. The remainder of the urethra situated in the penis is referred to as the spongy urethra.

The prostate is commonly described as the size of a chestnut, being approximately 3 cm in diameter. The prostate produces approximately a third of the volume of seminal plasma.

The bulbourethral glands, sometimes referred to as the cowper's glands, reside on either side of the membranous urethra and their ducts empty into the base of the spongy urethra. These glands secrete a small quantity of mucus prior to ejaculation which is believed to lubricate the urethra and to neutralize traces of acidic urine. The urethra then passes through the penis and terminates at the external urethral opening in the tip of the glans penis, from which semen is expelled during ejaculation (Jones, 1997; Marieb, 1995).

2.2 | Sexual response and ejaculation

The sexual response cycle for both sexes is dived into four distinct phases. The phases are the excitement phase, the plateau phase, the orgasmic phase, and finally the resolution phase. For males, the excitement phase typically occurs in response to any effective erotic stimuli and results in the extension and hardening of the penis. There are a number of other physiological effects which occur during this phase, but none that specifically relate to semen production.

In the plateau phase, the erection is maintained, while the bulbourethral glands and the prostate become enlarged. The urethral bulb, a portion of the spongy urethra at the base of the penis, also enlarges to three times its normal size. The bulbourethral gland produces a thick clear secretion, known as pre-ejaculate. The volume of pre-ejaculate is typically around a few drops but can vary up to a few milliliters.

Views on the presence or absence of spermatozoa in pre-ejaculate vary and outside of the forensic field, these discussions focus on the efficacy of the withdrawal method of contraception. A planned parenthood organization indicates that the withdrawal method, when practiced perfectly, is responsible for pregnancies in 4 out of 100 women, and in real life situations this practice is responsible for 22 pregnancies out of 100 women (Planned Parenthood 2019). This is supported by the Mayo Clinic who indicate that 28 out of 100 women who practiced the withdrawal method over a period of a year became pregnant (Mayo Foundation 2019). One small scale study indicated that pre-ejaculatory secretions do not contain spermatozoa and therefore cannot be responsible for pregnancies during coitus interruptus (Zukerman, Weiss, & Orvieto, 2003); however, another group found motile spermatozoa in 37% of pre-ejaculatory samples, but that same study also showed that sperm were found in either all of the repeat samples produced by a volunteer or none of the repeat samples produced by a volunteer (Killick, Leary, Trussell, & Guthrie, 2011). This suggests that pre-ejaculate may contain spermatozoa; however the presence or absence may be individual specific and may also be dependent on spermatozoa remaining in the vas deferens or urethra from previous ejaculations.

The orgasmic phase of the cycle occurs when the stimuli is sufficient to trigger the ejaculation reflex. Ejaculation is controlled by the parasympathetic nervous system. This reflex directs a specific sequence of contractions of the smooth muscles in the sex accessory glands and ducts. These contractions expel semen into the urethral bulb. Simultaneously the internal urethral sphincter contracts and closes. This is termed the emission stage of ejaculation and is immediately followed by the expulsion phase during which intense contractions of the penis expel semen from the head of the penis. This phase is also accompanied by generalized muscle contractions, increased heart rate, and elevated blood pressure.

The following resolution phase entails a general return to normal resting state. This includes a relaxation of muscle tension, lowered heart rate, blood pressure, and respiratory rate, as well as a loss of erection (Jones, 1997).

3 | SEMINAL FLUID

3.1 | Contents and contributions

The volume and sperm count of a “normal” ejaculate varies slightly between publications. The World Health Organization provides data for reference values of human semen characteristics. They provide a range of values for ejaculate volume of 1.6–5.5 ml, and containing 17–192 million sperm per milliliter for the general population (Cooper et al., 2009).

The contribution to semen volume from the testes, being primarily spermatozoa, forms less than 5% of total ejaculate volume. The spermatozoa are tiny, streamlined motile cells containing a package of haploid DNA with very little cytoplasm or nutrients. Accordingly, they rely on the other constituents of semen to facilitate much of their function. The epididymis contributes a very slight amount being mainly carnitine and glycerylphosphocholine. Both carnitine and glycerylphosphocholine are involved in providing nutrients to the sperm.

The seminal vesicles provide approximately two-thirds of the volume of semen. This contains high levels of fructose which is used as the primary energy source for the spermatozoa. This contribution also contains amino acids, ascorbic acid, flavins, enzymes, fibrinogen, phosphocholine, and prostaglandins. The prostaglandins have a role in decreasing the viscosity of cervical mucus and inducing a reverse peristalsis in the female reproductive tract, both of which facilitate sperm transport. Fibrinogen encourages semen to coagulate shortly after emission, presumably to aid the retention within the vagina.

The prostate provides approximately one-third of the volume of semen. The prostate contribution to semen includes buffers, citric acid, zinc, enzymes (including fibrinolysin), prostate-specific antigen (PSA) and acid phosphatase. Spermatozoa are sluggish in an acid environment. Semen has a pH of 7.2–8.0 and citrate has been suggested as the source of semen's high buffering capacity which help maintain the slightly basic pH, hence maintaining sperm motility in the usually acidic vaginal environment. Zinc assists in stabilizing chromatin within the cell and zinc deficiency is implicated in lowered fertility due to increased sperm fragility. Fibrinolysin liquefies the clotted semen after a period in the vagina to facilitate sperm movement and transport.

The bulbourethral gland secretion volume is usually quite small, and neutralizes traces of acid urine in the urethra and provides lubrication, both in the urethra and for the head of the penis.

3.2 | Medical issues and the effect on seminal fluid constituents

There are many medical conditions relating to male fertility that are reflected in the morphology and numbers of sperm in an ejaculate. Male infertility rates vary internationally but are in the range of 2.5–9.4% of the population. Oligospermia is defined as a low sperm count (<15 million/ml) and is a common indicator of male infertility. Azoospermia is the complete lack of sperm in the ejaculate and occurs in approximately 5% of infertile males.

A number of medical or surgical factors may also have an influence on the expected contents of semen. A vasectomy is a medical procedure in which the vas deferens is severed just above the testes and tied off. Although the testes still continue to produce sperm, they are prevented from exiting the body during ejaculation. These blocked sperm die and are reabsorbed into the body. Given the minimal contribution to the ejaculate from the testes, the semen volume and the physical experience of ejaculation remain largely unchanged. Some sperm may exist within the remaining sexual accessory ducts and glands so it may take a few months for a vasectomized male to have a zero sperm count.

Testicular cancer is relatively rare, occurring in around 1 in every 20,000 males. The most common symptom of testicular cancer is a painless solid mass in the testes. When caught early, the testicular cancer is easily remedied with the removal of the testes and radiation treatment. Erectile dysfunction, reduced intensity of orgasm and perception of reduced semen volume are common after-effects of this procedure. Provided one testis remains, the semen will still contain spermatozoa and the patient can still be fertile.

Prostate cancer is the second most prevalent cause of cancer-related deaths for males. Other than by a digital rectal exam prostate cancer can also, and more effectively, be detected by monitoring for elevated levels of a PSA in the blood. Elevated acid phosphatase levels in the blood may also be diagnostic. When caught early, prostate cancer survival rates are very good; however surgical removal of the prostate and the seminal vesicles (radical prostatectomy) as well as radiation therapy are common treatment options. Erectile dysfunction and incontinence are common after-effects of this procedure. Also, the vas deferens is severed which effectively functions as a vasectomy and prevents any spermatozoa being present in the ejaculate. Although normal sex and ejaculation is still possible post-surgery, the removal of the prostate and the seminal vesicles means that there is no seminal fluid associated with the ejaculate, otherwise known as a “dry” ejaculate.

4 | FORENSIC TESTS FOR SEMEN

From the earliest forensic tests in cases of sexual assault, the detection of spermatozoa to confirm the presence of semen has been the irrefutable gold standard; however, there are a number of reasons why spermatozoa may not be detected. Any semen involved may not have sufficient numbers of spermatozoa to be readily detected, or in fact contain no spermatozoa at all. There are also a multitude of potential issues relating to transfer and persistence relevant to specific case circumstances that

could impact on the likelihood of finding spermatozoa. So, other chemical tests were needed to locate potential semen stains, or alternatively indicate that semen may be present, for subsequent confirmation of the presence of semen. Presumptive screening tests for semen are qualitative so a positive test result is indicative, but not confirmatory, for the presence of semen.

4.1 | Presumptive tests for semen

4.1.1 | Crystal tests

At the end of the 19th century, Florence introduced a crystal test for semen using a reagent containing potassium iodide and iodine (Gaensslen, 1984). The test was typically performed on a microscope slide, with the sample suspected of containing semen being placed on the slide under a cover slip, and the reagents then being allowed to diffuse under the coverslip. The characteristic brown choline periodide crystals could then be observed under a microscope as they formed at the interface of the two liquids. The crystals form in the presence of choline—phosphocholine occurs at high levels in semen, originating primarily in the seminal vesicles. It has been suggested that a prostatic enzyme was required to hydrolyze the phosphocholine to liberate the choline.

The Barberio test was performed in a similar way but utilized picric acid solutions to target the presence of spermine to form spermine picrate crystals. Historically there has been disagreement on whether a negative result for a crystal test could confirm the absence of semen or not, and likewise whether a positive result should be considered confirmatory or merely presumptive; however, some would suggest the Florence crystal test still has a place in the repertoire of forensic tests with its specificity exceeding commercially available PSA tests (Hardinge, Allard, Wain, & Watson, 2013).

4.1.2 | Alternative lights sources

The ability to locate an otherwise nonvisible semen stain reliably and accurately without having to handle an item excessively or without having to use chemicals would be the ideal forensic screening test. To achieve this, attempts have been made to capitalize on the property of semen to fluoresce over a relatively broad spectrum. Fluorescence is a form of luminescence whereby a material absorbs a particular wavelength of light and simultaneously re-emits it at a lower wavelength. Flavins and choline present in semen are the prime contenders for the ability of semen to fluoresce under UV light.

A relatively simple UV lamp, known as a Wood's lamp, has been traditionally used in the medical field to screen for semen. The wood's lamp typically emits long wave ultra violet (UV-A) light in the region of 315–400 nm. Unfortunately, the Wood's lamp has been shown to be unreliable. One study claims that of 29 semen samples from 29 different donors, none demonstrated any appreciable fluorescence when illuminated with a standard Wood's lamp at 360 nm (Santucci, Nelson, McQuillen, Duffy, & Linakis, 1999). Also, UV-A lamps have traditionally been considered safe and were routinely used as tanning lamps to facilitate melanin production, but more recently have been shown to contribute to skin cancer via indirect DNA damage. Interestingly, short wave ultra violet (UV-C) lights in the region of 265 nm can be used to sterilize biological laboratories as the light damages the DNA of microorganisms.

Many alternative light sources (ALS) have been developed for use in the screening of biological fluids. The excitation spectrum for semen has been determined at between approximately 300–480 nm, and the emission spectrum stretches out over 400–700 nm. Correspondingly, many procedures have been proposed that utilize a combination of specific excitation bands and colored filters, with the user wearing different colored goggles, to visualize semen stains (Stoilovic, 1991).

Although many forensic laboratories and police departments are advocates for ALS, it is widely acknowledged that it has some significant limitations. White or lightly colored fabrics containing optical brighteners can demonstrate a high background fluorescence, which can mask any fluorescence from a semen stain. Conversely, darkly colored fabrics have proven difficult when attempting to locate semen stains via an ALS method. Highly absorbent fabrics also proved difficult when attempting to detect semen, especially if the semen was diluted, such as often is the case in actual forensic casework where semen is often presented as a component of vaginal drainage captured in underwear. To complicate matters further, there are a range of commonly encountered commercial products, such as soaps or hand creams that can also fluoresce producing a similar spectrum to semen. With relatively high potential for false negatives and false positives, ALS should not be used in isolation for semen detection. It may have some value when used in conjunction with other testing methods, such as the AP tests, as a tool to target areas for further examination (Kobus, Silenieks, & Scharnberg, 2002); however, for operational forensic laboratories, it is often difficult to justify utilizing an ALS technique if that technique then needs to be checked by a second chemical technique.

4.1.3 | Acid phosphatase

The most routinely used screening test for the presence of seminal fluid staining on intimate swabs, garments, or at crime scenes is the acid phosphatase (AP or AcP) test, or Brentamine blue test. This test utilizes a reagent containing a naphthyl phosphate salt in solution with a stabilized diazonium compound. In the presence of acid phosphatase, the naphthyl phosphate is hydrolyzed to produce α-naphthol which in turn couples to the Brentamine Fast Blue to form an intensely colored purple azo dye.

Kind (1958) is widely credited as developing the current form of the screening technique, and the robustness of the test has meant that the process has gone largely unchanged. This work was likely built upon previous work which demonstrated that semen contained significantly higher levels of acid phosphatase than other biological fluids, or other commonly encountered liquids or beverages (Gaensslen, 1984). It was shown that the minumum level of AP observed in semen still exceeded those found in other biological fluids by nearly 200 times. These tests provide good estimates of the relative AP response for semen in comparison to other substances when performing presumptive chemical testing.

Research into other testing regimes, primarily in the 1970s and 1980s, produced a plethora of alternative reagent options, however the Brentamine blue version remains the most commonly used. Variations in the technique often reflect a laboratories preference in terms of available resources, safety, sensitivity, and specificity.

The most common application of the AP test is often referred to as a "press test" and is undertaken by placing a moistened piece of filter paper, or similar material, onto the item to be tested, followed by pressing the paper firmly. A small quantity of any semen present is solubilized and absorbed onto the filter paper. The filter paper is then removed, the AP test reagent is applied to the paper and a colored response is obtained in the presence of AP. This is known as chemical mapping as the colored test results can be "mapped" back to locations on the test items. This test is also referred to as an indirect test as the reagent is not applied directly to the test item and may require the test paper to be relocated on the item to accurately indicate the locations of the positive response. The direct test approach requires the application of the AP reagent directly to the item and observing the response. This approach is much more suited to items where the testing can be easily targeted and a colored response easily observed, such as on swabs, however it has the added complication of adding a chemical reagent to the sample which could potentially complicate downstream analyses.

The range of time taken for a colored response to develop, and therefore be used as an indicator for the presence of semen, has been questioned. The time and color of the response for neat semen used as positive controls is often just a few seconds. Traditionally, an observed cut-off time of 2 min for recording a positive response has been used by many laboratories, with response time at 2 min or longer often considered to be more indicative of a vaginal acid phosphatase (VAP) response. There are competing objectives of sensitivity and specificity with some researchers arguing that times of up to 10 min or longer (Lewis, Jones, Baxter, Siemieniuk, & Talbot, 2012; Redhead & Brown, 2013) should be considered, while others suggest reducing the time to as little as 15 s would increase the specificity and reduce the likelihood of false positives (Peonim et al., 2013).

It is widely accepted that acid phosphatases can be present at lower levels in a range of other bodily fluids which can give a response to this test, as well as some vegetable juices, specifically cauliflower, and some contraceptive creams. Therefore, this test is only an indicative, or presumptive, test for the presence of semen (Hooft & van de Voorde, 1994). Generally, non-semen reactivity is significantly slower and provides less intense, and often a different colored response than semen, for example, some teas can provide a rose colored response to the AP reagent.

The most significant issue with AP test for semen is the potential overlap between very low concentrations of seminal acid phosphatase and very high concentrations of vaginal acid phosphatase. With respect to sexual assault investigations, the other bodily fluid most realistically expected to be encountered is vaginal secretions. The term vaginal secretions does not relate to a single secretion from a definable source and is used to describe collectively any material which is recovered on vaginal swabs or washings (Sensabaugh, 1979). It has been shown that seminal AP activity on vaginal swabs does decline significantly for samples collected at 24 hr post coitus and is unlikely to be able to be detected in samples collected after 48 hr. This predictable break down of seminal AP can, in some circumstances, assist in determining, or differentiating, interpretations around the time since intercourse.

The AP reactivity on items is considered to be relatively stable once the seminal fluid stain is dried and positive AP results can be obtained from semen stains that are months or even years old (Nakanishi, Hara, Takahashi, Takada, & Saito, 2014); however, AP reactivity does deteriorate under heat or moisture so unfavorable environmental or storage conditions can have a detrimental effect. The washing of items, such as clothing, containing semen stains should effectively remove all of the water soluble components of semen; however, some of the physical components of semen, specifically in the form of spermatozoa, may be retained within the fabric of a garment or redistributed on that garment or other garments it has been washed with

(Crowe, Moss, & Elliot, 2000). This in itself can create complications in interpreting semen findings in domestic cases involving cohabitation.

Once a presumptive semen stain has been detected, it can then be sampled for further testing to confirm the presence of semen, and potentially to provide a DNA profile to determine who that semen could have originated from.

4.2 | Confirmatory tests for semen

4.2.1 | Microscopic identification

The microscopic identification of spermatozoa remains the most widely accepted method for confirming the presence of semen. Spermatozoa are only found in semen therefore finding a single confirmed spermatozoon head is sufficient evidence to say that the sample contains semen (Saferstein, 2005). This method was first introduced into forensic science in 1839 by Henri-Louis Bayard.

The human spermatozoon is a flagellated cell that is made up of a head, neck, mid-piece, and a tail. Its total length can range from 40 to 250 μm, with the oval-shaped head size 4–5 μm in length and approximately 2–3 μm wide (Jones, 1997). The sperm head contains a haploid nucleus which is surrounded by a nuclear membrane. A cap fits over the sperm head called an acrosome which contains enzymes important in the penetration of the ovum. The mid-piece of the spermatozoon contains mitochondria to generate energy for tail movement. The mid-piece and sperm tail provide a propulsive force allowing for locomotion.

It is known that spermatozoa are generally robust and well preserved in dried stains and when in cadavers, however they are lost easily from the body cavities of live victims. This loss maybe the result of drainage, activity, and bathing/washing (Davies & Wilson, 1974). The morphological characteristic of a sperm cell degrades over time inside the vagina and this can lead to loss of quality and a reduction in sperm numbers (Casey et al., 2017). The sperm tails are also easily dissociated from the head and mid-piece. Intact spermatozoa are not often recovered in forensic samples; however the cell can still be visually identified under a microscope through the observation of the sperm head. Although the presence of spermatozoa in the vaginal cavity provides evidence of sexual intercourse, it may also be necessary to determine the approximate time of sexual activity. The loss of the tail is considered an indication of seminal degradation and can be used to assist in providing an indication of the time since intercourse (Collins & Bennett, 2001; Davies & Wilson, 1974). Time since intercourse can be estimated based on the number of intact spermatozoa (tails intact) and the number of sperm heads (no sperm tails present).

Several histological stains are available that allow for spermatozoa to be visualized against a background of other cells such as epithelial cells and/or debris. In alleged sexual assault casework, vaginal samples are commonly encountered. Not surprisingly, the vaginal secretions in these samples contain epithelial cells that need to be differentiated from spermatozoa within the sample. Most histological stains will stain the nuclei head of a spermatozoa the same color as the epithelial cell nuclei. A desired stain for spermatozoa would allow for the easy differentiation of the epithelial cells and spermatozoa. Christmas tree stain was developed for forensic applications in 1969 and remains a popular choice in forensic laboratories (Saferstein, 2005). This stain incorporates nuclear fast red and picroindigocarmine for staining in a two-step process. This will stain the nuclei of the epithelial cells and the sperm heads red while staining the cytoplasm of the epithelial cells green. This allows for easy visualization of bipolar sperm heads against a background of epithelial cells.

Hematoxylin and eosin (H&E) is another commonly used stain. Hematoxylin is a nucleic acid that stains the nuclei purple while eosin stains the sperm tails and the cytoplasm of the epithelial cells pink.

The main disadvantage of microscopic examination occurs if the donor or sample lacks spermatozoa. Many factors will influence the recovery of spermatozoa including but not limited to azoospermia, oligospermia, condom use, coitus interruptus, penetration with no ejaculation, and a vasectomy.

Alternative staining methods have been proposed to tackle this issue, such as fluorescent in situ hybridization (FISH). In situations where spermatozoa are not likely to be present it is beneficial to consider other male cells within a samples. Epithelial cells can be deposited during a sexual assault, for example, during penetration and/or ejaculation. The FISH staining technique incorporates different colored DNA probes for particular parts of chromosomes in the nucleus. When coupled with laser microdissection, FISH can be an effective way to obtain DNA profiling information from specific cell groups that have been fluorescently labeled, such as those male cells containing X and Y chromosomes (Lynch, Gamblin, Vintiner, & Simons, 2015).

As forensic science continues to evolve, new methods for sperm identification are introduced. SPERM HY-LITER™ (Independent Forensics Illinois, Lombard, IL) was released as the first commercially available immunofluorescence detection kit for the screening and identification of sperm cells from sexual assault evidence (Miller et al., 2011).

This method was designed to alleviate the laborious process of microscopic identification of spermatozoa with histological stains. This kit incorporates an Alexa 488 fluorescently tagged monoclonal antibody, which targets an antigen on the nuclear membrane of sperm cells (Westring et al., 2014). It also contains 4′-6-diamidino-2-phenylindole (DAPI) for simultaneous viewing of all cell nuclei. The specific nature of the labeling means that sperm cells can be readily identified in the presence of other nontarget cell types. SPERM HY-LITER™ techniques do however, require a microscope that is fitted with a fluorescein isothiocyanate filter. The staining protocol is also time consuming when compared to alternative histological staining techniques and can take 1–2 hr depending on whether the standard or the express kit is being used. Other limitations of the procedure include nonspecific staining, cost, and increased background fluorescence (Miller et al., 2011).

4.2.2 | Prostatic-specific antigen (PSA)

Forensic samples containing no spermatozoa or very low numbers of spermatozoa have always been problematic for confirming the presence of semen. The use of seminal fluid markers from the accessory sex glands provides an alternative method of testing (Saferstein, 2005). In forensic casework PSA, also known as p30, is the marker of choice.

Prostate-specific antigen was discovered in the 1970s, and is a serine protease which is produced by the prostate gland (Kearsey, Louie, & Poon, 2001). It is considered a useful semen marker for forensic samples due to its stability, immunogenicity, low-level detectability, and tissue specificity (Sensabaugh, 1978). It was initially thought the PSA was prostate specific; however it has been detected in other tissues and body fluids including nipple aspirate fluid and breast milk from females (Graves, 1995; Saferstein, 2005; Yu & Diamandis, 1995). Male urine has also been reported to contain traces of PSA. However, this could be due to drainage from the prostatic ducts and the periurethral glands into the urethra. PSA levels above 4 ng/ml have also been observed in the blood of patients with prostate cancer (Kearsey et al., 2001).

Sensabaugh (1978) concluded that PSA was stable in semen stains and present in semen at sufficient levels to be detected in stain extracts. The level of PSA in normal human semen is found to range between 0.24 and 5.5 mg/ml therefore even at the low end of the range, PSA can be easily detected. It was also noted that PSA persistence in the vagina cavity is parallel to that of acid phosphatase activity (Sensabaugh, 1978). Although the acid phosphatase method is very easy and quick to carry out it lacks tissue specificity and is primarily used in forensic casework as a screening tool.

Several methods have been developed to detect PSA including enzyme-linked immunosorbent assay (ELISA), Ouchterlony double diffusion, and cross over electrophoresis (Hochmeister et al., 1999). These techniques are time consuming, some lack specificity and are not practical for forensic laboratories. Commercial membrane test strips based on immunodiffusion are now readily available. These include the Seratec® PSA Semiquant (Seratec Diagnostica, Gesellschaft fuer Biotechnologie mbH, Göttingen, Germany), and the One Step ABAcard® (Abacus, Diagnostics, West Hills, CA).

Commercially available kits such as the ABAcard® test have been shown to have a sensitivity of 4 ng/ml (Kearsey et al., 2001). This is an immunochromatographic membrane test that contains a disposable plastic card with a sample well and a viewing chamber. This test is very straightforward and takes 10 min to perform after the stain is extracted. Once the sample is added to the sample well, a mobile monoclonal immunoglobulin complexes with PSA. This complex diffuses along the membrane toward the test area containing immobilized polyclonal anti-PSA antibody. The immobilized antibody captures the antigen–antibody complex and a pink line is formed due to the conjugated pink dye antigen–antibody complex being concentrated to this test region. The sample continues to migrate down the absorbent device which also contains a control band which is loaded with immobile anti-immunoglobulin polyclonal antibody, this attaches to the bound or unbound monoclonal anti p30 antibody forming a second pink band. When a sample contains PSA then two pink bands will form, when a sample is negative or contains less than 4 ng/ml of p30 only the control band will form. Detailed validation studies have been conducted to show that the ABAcard p30 test is a suitable replacement for the ELISA (Hochmeister et al., 1999; Kearsey et al., 2001).

This test has been found to detect seminal fluid in a variety of human semen stain extracts and is able to detect seminal fluid diluted 1:50,000 to 1:1,000,000 (Hochmeister et al., 1999, Kearsey et al., 2001).

During validation of the ABAcard PSA test, Kearsey et al. (2001) found no detectable PSA levels in saliva, blood, or female urine. They also noted that p30 could be detected on vaginal swabs up to 24 hr post coitus but generally not by 48 hr post coitus. Overall, in this validation only male urine was found to give a false positive result.

When using a commercially available immunochromatographic membrane test a negative result may indicate the absence of the antigen, or that the antigen is below detectable levels or that there is an excess of the antigen in the sample. The latter is referred to as the high dose hook effect. The manufactures of the ABAcard state the high dose hook effect has been observed at concentrations of PSA greater than 50,000 ng/ml. It has been recommended if a high does hook effect is suspected then a dilution of 1:100 should be considered to remedy this (Hochmeister et al., 1999).

The ABAcard test allows for samples to be prepared for p30 testing and then subsequently processed for DNA analysis with minimal sample wastage. This has enabled the technique to be readily implemented into routine forensic casework.

The RSID™-Semen test kit (Independent Forensics, Hillside, IL) is another commercially available kit for the detection of semen semenogelin, this kit does not detect PSA or p30. Semenogelin is a protein that is produced by the seminal vesicles and is responsible for the clotting associated with ejaculation. Boward et al. (2013) and Pang and Cheung (2007) compared the RSID-Semen™ test to the ABAcard® p30 test card. Boward et al. (2013) found that they were both specific at identifying semen however the ABAcard® p30 test outperformed the RSID™ Semen test in sensitivity, cost per test, and ease of protocol.

4.2.3 | Messenger RNA

Most of the new methods being developed to identify semen and other body fluids involve RNA markers. Messenger RNA (mRNA) can provide cell and tissue specific information that can be utilized for body fluid identification. It was previously thought that RNA was unstable and degraded rapidly making it unsuitable for forensic application. It has since been shown to be stable in a dried biological stain (Juusola & Ballantyne, 2005). Juusola and Ballantyne (2005) demonstrated that RNA can also be recovered in sufficient quantity and quality for analysis and in 2005 they introduced a multiplex reverse transcription-polymerase chain reaction (RT-PCR) method for body fluid identification of forensic samples. RT-PCR coupled with capillary electrophoresis is the most routinely used mRNA method in forensic science.

In order for an mRNA method to be a valuable forensic tool for the detection of semen it must be able to identify seminal fluid in both the presence and absence of spermatozoa. Core markers have been used for multiplex design including protamines 1 and 2 (PRM1/2) for spermatozoa and transglutaminase 4 or semenogelin 1 (SEMG1) for seminal fluid. In designing an mRNA multiplex system Fleming and Harbison (2010) focused on PRM2 for spermatozoa and transglutaminase 4 for seminal fluid. In this research, they were able to identify seminal fluid without spermatozoa from 1 μl of seminal fluid. The multiplex created was able to identify circulatory blood, menstrual blood, semen, and saliva while allowing for coextraction of the nuclear DNA for profiling.

It has been argued that an mRNA approach has several advantages over the biochemical methods, such as greater specificity, decreased sample consumption, and compatibility with DNA extraction (Fleming & Harbison, 2010). It has been described as the definitive method for body fluid identification (Fleming & Harbison, 2010). There are now methods that allow for the coextraction of the DNA from the same sample reinforcing its application to routine casework (Fleming & Harbison, 2010).

5 | EMERGING TECHNOLOGY

As new research is conducted in the area of semen identification for forensic application it is likely that the future will incorporate a combination of conventional and innovative methods.

5.1 | Sperm tracker

SPERM TRACKER™ was released in 2017 as a commercial product specially designed for the detection of semen traces. This product promises the specificity of the acid phosphatase reaction with improved ease of use while also being nontoxic (Borges et al., 2017).

Sperm Tracker is sold in sheets that can be used on larger items, however it requires the room to be dark and needs a UV light source for visualization. The greatest advantage of this method is the ability to leave the paper in contact with the item of evidence while it is screened for semen, making it a direct testing method, whereas AP screening requires the paper to be removed for testing. However, if the item of evidence auto fluoresces under UV light the Sperm Tracker paper will need to be removed to allow for visualization. It has been found to outperform the mini-CrimeScope technique but its benefits over conventional acid phosphatase screening are yet to be explored.

5.2 | Digital background reduction in florescent imaging

Background fluorescence is one of the limitations involved in the fluorescent imaging of semen stains. With developments in digital photography, there is increasing scope for the ability to capture a suite of images using narrow wavelength band filters

and to process this information using background correction algorithms. This allows for the reduction of interference in the images and can optimize the desired fluorescence either from the semen stain, or from a chemically enhanced semen stain.

5.3 | Raman spectroscopy

Raman spectroscopy is not a new technique and has been discussed as a body fluid identification tool for many years. It is claimed that Raman spectroscopy can be used to differentiate body fluids based on their different chemical compositions. Typical Raman spectroscopy gives a specific vibrational signature of the sample being measured based on the energy of scattered light. The identification of a sample as being semen is based on the application of a calculated spectroscopic signature. However, the resulting spectrum is complex and advanced statistical analysis is required. This method requires no sample preparation, no reagents, minimal sample requirement, is nondestructible and is portable. It has also been said to be confirmatory and can be used to identify body fluids at crime scenes (Virkler & Lednev, 2009). Portable devices have now been developed for measuring other substrates such as white powders and narcotics. These devices could potentially be expanded to body fluid identification in the field (Virkler & Lednev, 2009). Although Raman spectroscopy appears to work under controlled laboratory conditions, more research is needed to determine its capability in casework scenarios.

5.4 | Mass spectrometry

Advances in mass-spectrometry instrumentation and software have seen its use in forensic body fluid identification being explored (Yang, Zhou, Deng, Prinz, & Siegel, 2013). Mass spectrometry uses the knowledge that protein expression is linked to its function. Therefore, a body fluid can be identified by its functional protein markers. Mass-spectrometry based techniques have been found to detect body fluid biomarker proteins accurately and with high sensitivity (Kamanna, Henry, Voelcker, Linacre, & Kirkbride, 2016; Yang et al., 2013). Methods have also been developed to allow for the positive identification of a broad range of body fluids for practical use in forensic casework (Kamanna et al., 2016).

5.5 | DNA methylation profiling

Recent advances in the field of epigenetics have determined that it is possible to differentiate DNA methylation profiles for different types of cells or tissue. DNA methylation typically involves the addition of a methyl group to the 5′ position of a pyrimidine ring of cytosine in CpG nucleotides. This DNA modification plays an important role in controlling and modulating genome expression and the repression of gene transcription which in turn allows cells to maintain different characteristics (Lee, Park, Choi, Yang, & Shi, 2012).

BOX 1 SPERM NUMBERS AND THE DECLINE OF MALE FERTILITY

The decline of sperm numbers over the last 50 years has been widely reported, sparking much debate, and some controversy in the arena of reproductive medicine. A recent comprehensive meta-analysis of the research indicates that sperm counts within Europe, North America, and Australia have declined by 50–60% between 1973 and 2011, with no evidence of leveling off in recent years (Levine et al., 2017). Much of the debate focuses on whether this is a result, at least in part, of in-utero effects on the male fetus due to man-made environment factors, and whether we will ultimately be the agents of our own demise, or if in fact there is sufficient data to make any interpretations at all (Toppari et al., 1996). The global rates of infertility are based on self-reporting and consequently vary greatly across countries and cultures. Europe, North America, and Australia were considered as regions where males would likely feel comfortable addressing infertility issues with a physician which would allow the problem to be addressed and reported statistically. They reported rate of male infertility, respectively, as 7.5, 6, and 8% (Agarwal, Mulgund et al., 2015). The rate of infertility in many less developed countries were unknown, but were estimated to be higher. At least 30 million men worldwide were considered to be infertile. The main message of these studies was that male infertility was a global health issue and that further research is essential to truly understand its prevalence, magnitude and whether the causes can be mitigated.

Sets of epigenetic markers have been developed to differentiate between cell types, specifically blood, menstrual blood, saliva and, or of particular interest, semen. Validation studies on specific markers are showing promise in sensitivity, specificity, and application to case type samples and mixtures. However, further validation work is still required to confirm the robustness and reliability of the methodology and interpretations (Lee, Jung, Lee, Yang, & Shi, 2016; Silva et al., 2016).

A key feature of DNA methylation identification of forensic samples is that it is compatible with current DNA profiling and with massive parallel sequencing and has the potential to identify not only an individual's DNA profile but the cell types that profile was obtained from (Richards, Patel, Stevenson, & Harbison, 2018). This technique could prove hugely beneficial for many sexual assault investigations and provide a key link that has, until now, been missing (Box 1).

6 | CONCLUSION

The objective of this article was to provide an overview of the constituents of seminal fluid and the tests that have been employed to detect this body fluid in forensic investigations. In alleged sexual assault investigations there are standard chemical screening tests to allow for the presumptive detection of semen. Further measures can then be taken to confirm or support the presence of semen either through microscopic examination or using other methods outlined in this article. This seminal fluid can then be linked to an individual through DNA profiling.

It would appear that although new technology is being investigated, the traditional methods of detection and identification remain the most widely used. Microscopic identification of spermatozoa is still regarded the most reliable confirmation that the sample contains semen. However, this does not take into consideration azoospermia, oligospermia, condom use, or those samples from vasectomized males. Microscopy is also not a screening tool that can be applied at crime scenes or used to search large items.

Abnormally low sperm count or no sperm producing males can still produce detectable amounts of seminal fluid that contains both prostatic gland and seminal vesicle secretions. Alternative methods have been developed to assist in detecting semen in these situations. It is also noted that not all cases of sexual assault result in the deposition of semen therefore alternative methods have been developed which can detect the male Y chromosome in epithelial cell mixtures. The use of mRNAs to identify body fluids at a molecular level remains promising, yet still appears in its early stages for forensic applications.

Forensic scientists may also need to take into consideration what impacts the apparent continuing decline in sperm numbers may have on semen identification. Most of the established methods for semen identification are aimed at normal healthy semen. This may see new innovative methods being introduced to the forensic arena.

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CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

AUTHOR CONTRIBUTIONS

Rian K. Morgan-Smith: Conceptualization; writing-original draft; and writing-review and editing-Equal. **Amelia P. Gamblin**: Conceptualization; writing-original draft; and writing-review and editing.

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